

QINGYUAN, China

WMO: 542590

Lat: 42.0736N

Lon: 124.8660E

Elev: 755

StdP: 14.30

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-11.4	-20.8	1.8	-13.5	-17.1	2.2	-9.3	16.2	20.6	14.0	19.0	0.9	250	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	89.7	71.3	87.2	71.1	84.7	70.2	76.4	84.4	75.0	82.1	73.6	80.1	6.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.2	131.5	80.0	72.9	125.6	78.8	71.6	120.1	77.8	40.6	84.6	39.2	82.1	37.9	80.2	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	13.8	11.7	-20.9	93.7	4.4	3.1	-24.0	95.9	-26.5	97.7	-29.0	99.4	-32.1	101.6		
(5)			-21.1	78.2	4.4	1.7	-24.2	79.4	-26.8	80.4	-29.3	81.4	-32.5	82.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	7.4	16.3	31.4	47.2	59.7	69.4	75.1	72.7	62.0	46.9	28.9	12.3		
	DBStd	24.61	8.62	9.81	8.99	8.51	6.33	5.01	3.92	4.83	6.21	8.06	10.65	9.52		
	HDD50	4963	1320	944	579	151	6	0	0	0	3	154	637	1169		
	HDD65	8344	1785	1364	1043	536	187	17	1	6	127	560	1084	1634		
	CDD50	2862	0	0	1	67	306	582	778	705	361	59	3	0		
	CDD65	768	0	0	0	2	22	149	313	246	36	0	0	0		
	CDH74	6499	0	0	0	49	441	1396	2471	1795	337	10	0	0		
	CDH80	2020	0	0	0	6	99	469	848	552	46	0	0	0		
	WSAvg	3.9	3.0	3.8	5.1	5.7	5.3	4.1	3.7	3.3	3.3	3.7	3.6	2.9		
	PrecAvg	31.9	0.3	0.4	0.7	1.9	2.7	4.6	7.7	7.4	3.0	1.9	1.0	0.4		
(15) Precipitation	PrecMax	49.5	1.3	1.4	2.3	3.9	5.2	10.6	17.8	19.3	6.7	5.0	2.4	1.1		
	PrecMin	20.4	0.0	0.0	0.0	0.6	1.0	2.4	1.0	1.3	0.7	0.2	0.2	0.0		
	PrecStd	6.8	0.3	0.4	0.5	0.9	0.9	2.0	3.5	4.2	1.5	1.2	0.6	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	47.4	62.5	80.3	87.7	92.6	94.0	92.1	84.3	75.1	61.3	42.0		
		MCWB	32.2	37.4	47.4	56.6	63.1	66.8	75.2	73.7	69.0	59.1	49.6	36.3		
	2%	DB	30.8	41.4	56.0	73.8	83.1	88.5	90.0	88.0	80.8	70.2	54.8	36.3		
		MCWB	26.8	33.9	42.6	53.6	60.5	66.3	73.7	73.7	65.6	56.1	45.9	32.3		
	5%	DB	26.9	37.9	51.5	69.5	79.4	85.3	87.4	85.6	77.8	65.8	50.1	32.2		
		MCWB	23.4	31.7	39.9	51.1	59.0	66.1	73.2	72.8	63.4	53.6	43.0	28.7		
	10%	DB	22.8	33.6	46.8	65.0	75.3	82.0	84.8	83.1	74.9	61.7	45.2	28.2		
		MCWB	19.8	28.8	37.3	49.0	57.6	65.6	72.5	71.4	62.2	51.5	39.7	25.2		
	0.4%	WB	33.2	38.8	48.4	59.9	66.6	72.0	79.1	78.0	71.8	61.4	51.1	37.1		
		MCDB	36.6	44.4	60.9	75.4	80.8	81.8	88.9	86.0	81.3	71.5	58.8	40.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.2	35.2	44.0	55.6	63.4	70.4	77.1	76.2	68.9	58.0	47.0	32.5		
		MCDB	30.5	40.7	53.6	69.5	77.2	80.2	85.8	83.8	75.9	67.0	53.3	35.7		
	5%	WB	23.6	32.0	40.8	52.7	61.3	69.1	75.6	75.0	66.7	55.4	43.8	28.9		
		MCDB	26.8	37.1	50.1	66.9	74.5	78.8	83.0	81.9	73.7	63.8	49.3	31.9		
	10%	WB	19.6	28.9	37.8	50.1	59.3	67.8	74.2	73.7	64.6	52.9	40.0	25.2		
		MCDB	22.7	33.8	46.3	62.4	71.5	77.4	81.1	79.7	71.6	60.1	44.9	28.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.8	24.6	21.4	24.4	24.3	20.7	16.6	17.5	23.3	24.2	20.4	22.4		
		MCDBR	27.0	26.4	28.0	32.7	32.4	28.0	21.8	21.4	27.0	29.9	26.1	24.1		
		MCWBR	22.8	19.1	16.1	15.6	12.5	9.3	7.7	8.2	12.3	15.7	17.0	19.7		
		MCDBR	26.2	24.1	26.3	27.8	25.3	20.2	17.1	16.5	21.5	26.0	22.8	23.7		
	5% WB	MCWBR	22.5	17.9	15.8	14.0	11.1	8.0	7.1	7.3	10.5	14.8	15.9	20.2		
(40) Clear-Sky Solar Irradiance	taub		0.366	0.424	0.496	0.585	0.637	0.695	0.661	0.580	0.507	0.525	0.443	0.379		
	taud		2.005	1.854	1.783	1.665	1.608	1.567	1.677	1.839	1.938	1.802	1.915	1.985		
	Ebn at Noon		230	234	233	223	215	203	209	222	228	199	198	211		
	Edh at Noon		37	51	62	75	82	85	76	62	53	54	41	35		
(44) All-Sky Solar Radiation	RadAvg		758	1050	1381	1565	1764	1642	1492	1474	1386	1024	696	608		
	RadStd		50	69	75	107	144	168	131	131	100	67	79	47		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page